VocaVision — Technology Stack & Architecture

Assistive AI System for Real-Time Color, Clothing & Food Recognition with Voice Output

GitHub: github.com/RAJ-A58/VocaVision | **Framework:** PyTorch (CUDA) + TensorFlow | **Python:** 3.13

1. Deep Learning & Machine Learning

Technology / Module	Role & Engineering Implementation
PyTorch 2.6.0 + cu124	Core framework for the Clothing Classifier v2. Implements MobileNetV2 architecture with a custom classification head.
CUDA 12.4 Acceleration	Hardware-accelerated training and inference on NVIDIA RTX 3050 Laptop GPU (6GB VRAM).
torch.cuda.amp	Automatic Mixed Precision (AMP) using FP16 tensors to accelerate training and reduce GPU memory consumption.
Data Augmentations	PyTorch torchvision transforms: RandomHorizontalFlip, RandomRotation, RandomAffine, and ColorJitter to bridge the domain gap.
TensorFlow / Keras 2.20	Framework powering the Food Classifier (MobileNetV2 fine-tuned on Food-101) and legacy clothing CNN fallback.
TensorFlow Datasets (TFDS)	High-performance data streaming, pipelining, and preprocessing for the Food-101 dataset.
Scikit-Learn (K-Means)	Unsupervised color centroid clustering to extract dominant color centers from segmented pixels.

2. Computer Vision & Image Processing

Component	Function & Technical Scope
OpenCV (cv2)	Real-time webcam video stream capture, frame resizing, format conversions, and contour drawing.
GrabCut Algorithm (OpenCV)	Foreground-background extraction isolating the clothing article from complex indoor backgrounds.
rembg & onnxruntime	Neural background removal using the lightweight U²-Net (u2netp.onnx, 4.5 MB) ONNX model.
HSV Color Space Mapping	Maps K-Means centroid BGR clusters into calibrated Hue-Saturation-Value boundaries with human-readable names.
Test-Time Augmentation (TTA)	Inference ensembling averaging predictions across multiple geometric transformations to improve accuracy by +2-3%.
Pillow (PIL)	Image type conversion layer mediating between OpenCV NumPy arrays, PyTorch Tensors, and Gradio interfaces.

3. Assistive Audio & Voice Synthesis (TTS)

Library / Tool	Application & Integration
pyttsx3 (SAPI5)	Native offline Text-to-Speech library utilizing Windows SAPI5 synthesizer for standalone desktop execution without internet.

Python Threading	Non-blocking, asynchronous speech dispatch (speak_async) allowing the camera stream to stay smooth during voice playback.
gTTS (Google Text-to-Speech)	Cloud-based TTS audio synthesis generating MP3 audio files for the web browser interface.

4. Web Interface & UI Framework

Framework	Deployment & Features
Gradio 6.0	Reactive web UI providing image upload, audio output playback, confidence charts, and segmented mask inspection.
Gradio Live Tunneling	Generates temporary public URLs (gradio.live) enabling remote browser testing and demonstration.
Hugging Face Spaces Ready	Designed for containerized cloud deployment with zero hardware dependency changes.

5. Development Environment, Hardware & Version Control

Resource	Specification
Language & Environment	Python 3.13 (64-bit) running on Windows 11 / PowerShell.
Hardware Accelerator	NVIDIA GeForce RTX 3050 Laptop GPU (6.4 GB VRAM) running Driver 596.21.
Git & Git LFS	Version control with Git Large File Storage tracking binary neural network weights (*.keras, *.pt).
Matplotlib & Seaborn	Generates training accuracy/loss progression plots and real-time horizontal confidence distribution charts.

6. Architectural Highlights & System Design

Architecture Pillar	Strategic Benefit
Hybrid Inference Engine	Coordinates PyTorch and TensorFlow models simultaneously in memory without cross-framework conflicts.
Hierarchical Decision Logic	Implements confidence margin thresholding (FOOD_MIN_THRESHOLD = 0.75, FOOD_MARGIN = 0.30) to eliminate false positives.
Mask-Restricted Color Analysis	Applies K-Means clustering strictly within GrabCut foreground masks, filtering out shadows and background colors.